

Data Sufficiency & Balance Risk Report

Impact Analysis and Detailed Mitigation Strategies for a Phased Location-Intelligence & SME Revenue Prediction Project

Project strategy in scope: Begin with **Option C — Location Intelligence without POS data** (Google Maps reviews, foot traffic estimates, population, competitor density, nearby POIs) to validate the concept and start generating data immediately. Once sufficient SME sales data has been collected through this process, transition to **Option B — SME POS-based forecasting** (Prophet / XGBoost Time Series) for higher-precision, business-specific revenue forecasting.

Scope note: This version of the report reflects a café-only SME targeting strategy — only café-type restaurants are collected and modeled, rather than mixing cafés with BBQ, Thai, Japanese, or other restaurant categories. As a result, the “unbalanced restaurant types” risk described in general SME modeling guidance does not apply to this project (see Section 4.3).

1. Why Data Sufficiency and Balance Matter

Every machine learning model — whether XGBoost, Prophet, or a Temporal Fusion Transformer — is only as reliable as the data it is trained on. Even a well-designed model architecture will produce misleading results if the underlying dataset is too small, too narrow, or unevenly distributed across restaurant types, locations, and time periods. This section documents exactly what goes wrong when data is insufficient or imbalanced, why it happens, what the practical consequences are, and — most importantly — how each problem can be prevented or corrected within the Option C → Option B roadmap.

2. Master Summary — Data Problems, Effects, and Risk Levels

Data Problem	Example	What Happens During Training	Effect on Prediction	Risk
Too few samples	1 restaurant × 24 monthly records = 24 rows	Model cannot learn stable patterns	Predictions fluctuate and are unreliable	Very High
Too many features vs. samples	24 rows, 20 features	Model memorizes training data (overfitting)	Accurate on training data, fails on new restaurants	Very High
Unbalanced restaurant types (N/A — scope is café-only)	Collecting only café SMEs by design, not mixing in BBQ / Thai / Japanese	Model trains on one consistent business type	Consistent accuracy within cafés; not built to predict other restaurant types	Low
Unbalanced locations	Mostly Bangkok, few rural areas	Learns city-specific patterns	Poor performance in rural locations	High

Data Problem	Example	What Happens During Training	Effect on Prediction	Risk
Short historical period	Only 6 months of data	Misses annual seasonality (holidays, tourism, festivals)	Cannot forecast yearly trends	Medium
Missing seasonal events	No New Year or Songkran data	Model never sees demand spikes	Underestimates holiday revenue	Medium
Missing values	Foot traffic missing for many months	Less usable information for learning	Reduced prediction accuracy	Medium
Noisy or incorrect data	Revenue entered incorrectly	Model learns false relationships	Unstable and inaccurate predictions	High
Data leakage	Using next month's revenue as an input	Model "cheats" during training	Excellent validation results but fails in production	Critical

Table 1: Overview of common data sufficiency and balance problems in SME revenue modeling.

3. Detailed Examples

3.1 Too Little Data (Single Restaurant, Monthly Records)

With only one restaurant reporting monthly revenue, two years of history produces just 24 rows of data — far too few for a supervised model like XGBoost, which typically needs the number of samples to be much larger than the number of features ("samples >> features").

Month	Revenue (THB)
Jan	850,000
Feb	870,000
Mar	910,000
...	...
Dec	950,000
Year 2 (12 more months)	...

Total: 24 rows. If the feature set includes 15–20 variables (foot traffic, competitor count, population, nearby offices, parking, restaurant size, restaurant type, average order value, etc.), the model has far too little evidence per feature to learn reliable relationships.

Note: Sections 3.2–3.3 illustrate the general, cross-category version of this problem for reference. As clarified in Section 4.3, this project's café-only scope avoids this specific risk by design — these examples remain useful context in case scope ever expands to other restaurant types.

3.2 Balanced Dataset (Good)

Restaurant Type	Number of Restaurants	Months	Total Samples
Café	25	24	600
Thai	25	24	600

Japanese	25	24	600
BBQ	25	24	600
Total	100		2,400

Even distribution across restaurant types lets the model see enough variety to generalize well to new, unseen businesses.

3.3 Unbalanced Dataset (Bad)

Restaurant Type	Number of Restaurants
Café	90
Thai	5
Japanese	3
BBQ	2

Result: the model mostly learns café patterns → prediction accuracy for cafés is high, but accuracy for BBQ restaurants is low. This is a classic class-imbalance problem.

3.4 Missing Seasonal Information

Month	Included?
Jan – Jun	■ Yes
Jul – Dec	■ No

If data only covers January–June, the model never learns patterns from back-to-school periods, high tourist season, Christmas/New Year, or year-end promotions — so forecasts for those months are likely to be inaccurate.

4. Advantages of Detected Problems + Detailed Solutions

The table below expands each problem from Section 2 with (a) why it is still valuable to know about it early (the “advantage” of catching it), and (b) concrete, actionable solutions that fit the Option C → Option B roadmap.

4.1 Too Few Samples

Risk level: **Very High**

Root cause: A single SME contributing only monthly records (e.g., 24 rows over two years) provides far too little evidence for a supervised model to find stable relationships between features and revenue.

Why catching it early helps:

- Identifying this early prevents wasted effort tuning a model (like XGBoost) that was never going to generalize on so little data.
- It redirects effort toward the right technique instead of the wrong one — saving months of trial and error before a thesis or demo deadline.

Detailed solutions:

- During Option C: rely on cross-sectional data (many locations, one snapshot in time) instead of waiting for a long time series from a single business — this avoids the small-sample trap entirely.
- Once real SME data starts arriving, pool data across multiple SMEs rather than modeling one restaurant in isolation (50 restaurants × 24 months = 1,200 rows, which is workable for XGBoost).
- If only one SME is available, switch to time-series methods built for small-sample forecasting: SARIMA, Prophet, or a Temporal Fusion Transformer only once enough history exists.
- Prefer daily or hourly POS granularity over monthly aggregates when it is available — daily data from one restaurant over two years yields about 730 rows, and hourly data yields roughly 17,520 rows.

4.2 Too Many Features Relative to Samples (Overfitting)

Risk level: **Very High**

Root cause: 24 rows with 15–20 features gives the model more “knobs” than data points, so it memorizes the training set instead of learning generalizable patterns.

Why catching it early helps:

- Recognizing this ratio problem early avoids shipping a model that looks perfect in testing but fails completely on any new restaurant — a critical issue to catch before a pilot or investor demo.

Detailed solutions:

- Apply feature selection / dimensionality reduction (e.g., correlation filtering, recursive feature elimination, or PCA) to keep only the most predictive variables.
- Use strong regularization in XGBoost (max_depth limits, min_child_weight, L1/L2 penalties, subsample and colsample_bytree < 1) to discourage memorization.
- Use k-fold or leave-one-restaurant-out cross-validation rather than a single train/test split, so overfitting is visible before deployment.
- Grow the sample size in parallel (more SMEs or finer time granularity) rather than only shrinking the feature set — both levers should move together.

4.3 Unbalanced Restaurant Types — Not Applicable Under Café-Only Scope

Risk level: **Low**

Root cause: This problem occurs only when multiple restaurant types are pooled together in an uneven mix (e.g., 90 cafés vs. 10 BBQ restaurants), causing the model to learn mostly café behavior and predict poorly for the underrepresented types. **Because this project intentionally targets café-type SMEs only, and does not combine other restaurant categories into the same dataset, this specific imbalance risk does not apply.** Every training example comes from the same business type, so there is no cross-category imbalance to correct.

Why catching it early helps:

- Removing restaurant-type mixing simplifies the modeling problem — one homogeneous business type means fewer confounding variables and a cleaner feature-to-revenue relationship to learn.
- It removes an entire category of stratification and class-weighting work described in the general case, freeing up effort for other higher-priority risks (sample size, seasonality, leakage).
- It makes results easier to defend: any accuracy figure reported applies uniformly to “cafés” rather than needing to be broken down and caveated per restaurant type.

Detailed solutions:

- Keep the intake pipeline scoped strictly to café-type SMEs during both Option C and Option B — do not opportunistically add BBQ, Thai, or Japanese restaurants into the same training set later, as this would silently reintroduce the imbalance problem.
- Still diversify —within— the café category itself (e.g., specialty coffee shops, dessert cafés, chain vs. independent, mall vs. street-front) so the model does not overfit to one narrow sub-style of café.
- Document the café-only scope explicitly in the methodology so it is clear the model is a café-specific revenue predictor, not a general restaurant model — this is a scope decision, not a limitation to hide.
- If the project later expands to other restaurant types, treat it as a new modeling phase: either train a separate model per type, or reintroduce the stratified-sampling and per-type evaluation practices from the general case at that point.

4.4 Unbalanced Locations

Risk level: High

Root cause: If most collected data comes from Bangkok with very few rural or provincial locations, the model learns predominantly urban dynamics.

Why catching it early helps:

- Identifying geographic imbalance early is directly useful for the Location Intelligence angle of the project — it is itself a finding worth presenting (“urban vs. rural prediction confidence differs”).

Detailed solutions:

- During Option C, deliberately diversify location sourcing using Google Maps / POI data across multiple provinces and city tiers, not just Bangkok.
- Include explicit location-context features (urban density tier, province, distance to city center) so the model can partly separate urban vs. rural patterns rather than conflating them.
- Segment validation metrics by geography (urban vs. suburban vs. rural) to quantify where confidence is lower and disclose this explicitly.
- If rural data stays scarce, present predictions for those areas with a wider confidence interval rather than a single point estimate.

4.5 Short Historical Period

Risk level: Medium

Root cause: Only six months of data cannot capture annual seasonality — holidays, tourist peaks, and festivals that repeat on a yearly cycle.

Why catching it early helps:

- Catching this early sets correct expectations: a 6-month pilot can validate the workflow and UX, but should not be presented as a full seasonal forecasting model yet.

Detailed solutions:

- Frame the Option C phase explicitly as a proof-of-concept for methodology, and only claim seasonal forecasting once at least one full annual cycle of data is collected.
- Where full history is unavailable, borrow seasonal patterns from public sources (tourism board statistics, retail sales indices, Google Trends) as proxy seasonal features.
- Prioritize backfilling historical POS data (even a rough monthly export) from new SME partners instead of only collecting data going forward.

4.6 Missing Seasonal Events

Risk level: Medium

Root cause: If the dataset never includes New Year, Songkran, or other demand-spike periods, the model has never “seen” these events and will systematically underestimate revenue for them.

Why catching it early helps:

- Recognizing this gap early prevents an embarrassing under-forecast being presented as reliable during exactly the periods that matter most to restaurant owners (holidays).

Detailed solutions:

- Explicitly add a categorical or binary “holiday/festival” feature (Songkran, New Year, Chinese New Year, long weekends) so the model can learn spike behavior once such data exists.
- Actively prioritize collecting data that spans at least one major holiday period, even if it means a short delay before finalizing the training set.
- Until enough holiday data exists, flag holiday-period forecasts as lower-confidence rather than presenting them with the same certainty as normal weeks.

4.7 Missing Values

Risk level: Medium

Root cause: Gaps such as missing foot-traffic readings for several months reduce the amount of usable signal the model can learn from.

Why catching it early helps:

- Identifying which fields are frequently missing early lets the team fix data collection pipelines before too much time and effort is invested in modeling on incomplete data.

Detailed solutions:

- Use appropriate imputation depending on the variable: forward-fill or seasonal interpolation for time-series gaps, median/mode imputation for cross-sectional gaps.
- Add a binary “was_missing” indicator feature alongside imputed values so the model can learn whether missingness itself is informative.
- Where possible, substitute a public proxy for the missing signal (e.g., Google Popular Times instead of a missing foot-traffic sensor reading).
- Track a data-completeness score per SME record and prioritize follow-up outreach with partners who have the most gaps.

4.8 Noisy or Incorrect Data

Risk level: **High**

Root cause: Incorrectly entered revenue or mislabeled records cause the model to learn false relationships between features and outcomes.

Why catching it early helps:

- Catching noisy data early avoids a much more expensive problem later: a model that appears to work but is quietly wrong in production, damaging credibility with SME partners.

Detailed solutions:

- Apply automated sanity checks (e.g., revenue cannot be negative, month-over-month change beyond a threshold gets flagged for manual review) as part of the intake pipeline.
- Cross-check self-reported SME figures against independent proxies (foot traffic estimates, review volume trends) to catch implausible outliers.
- Use robust modeling choices (e.g., Huber loss, winsorizing extreme values) so a small number of noisy records cannot dominate training.
- Keep a data-quality log per SME so noisy sources can be corrected or excluded transparently rather than silently degrading the whole dataset.

4.9 Data Leakage

Risk level: **Critical**

Root cause: Accidentally including future information (e.g., using next month's revenue as an input feature) lets the model “cheat” during training.

Why catching it early helps:

- This is the single most important issue to catch before publishing any result, because it can produce a model that looks excellent in validation yet fails immediately in real use — a serious credibility risk in a thesis defense or pilot with real SME partners.

Detailed solutions:

- Enforce a strict temporal split: training data must always end before the prediction target's time period; never validate with a random shuffle on time-series data.
- Audit every feature for a plausible “known-at-prediction-time” justification before it enters the model — if a feature could only be known after the fact, remove it.
- Use walk-forward (rolling-origin) validation for time-series forecasting rather than standard k-fold cross-validation.
- Re-check pipelines whenever new features are added, since leakage is often introduced accidentally during feature engineering rather than at the initial design stage.

5. How This Fits the Option C → Option B Roadmap

Phase	Data Source	Main Risks From Section 2–4	Mitigation Focus
Phase 1: Option C	Google Maps reviews, foot traffic estimates, population, competitor density, nearby POIs — no POS data required	Unbalanced locations; short historical period; low within-category (café sub-style) diversity	Diversify collection by area and business type; use public data as seasonal proxies; validate the modeling approach without depending on sensitive SME sales data
Transition Checkpoint	Enough SMEs have opted in and shared some sales data	Too few samples; too many features vs. samples	Check that samples ■ features before switching models; confirm at least ~20–50 SMEs or daily-level data from fewer SMEs
Phase 2: Option B	SME POS data (daily preferred over monthly) from participating restaurants	Missing seasonal events; noisy/incorrect data; data leakage	Anonymize and aggregate data; add holiday/event features; enforce strict time-based validation with Prophet / XGBoost Time Series

This phased approach directly addresses the core feasibility issue identified earlier: the hardest part of this project is not the choice of algorithm, but obtaining enough high-quality, balanced data from multiple businesses. Starting with Option C removes the initial data-collection bottleneck, builds credibility with SME partners over time, and creates a natural, evidence-based checkpoint for when to transition into Option B's higher-precision, POS-driven forecasting.